

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

Dec-25-0492

MCA-6102 (Database Management Systems)

MCA 1st (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from section A, B, C and D. Question no.9 is compulsory.

SECTION - A

1. Differentiate Relational Algebra and Relational Calculus? How a query is written in Relational Algebra and Relational Calculus? Explain with the suitable example. (10)
2. Describe three schema architecture with help of diagram. (10)

SECTION - B

3. What do you understand by trigger? Write syntax of different types of triggers used in SQL database along with suitable examples? (10)
4. What is normalization? What are advantages and disadvantages of normalization? (10)

SECTION - C

5. What are the various properties of Transaction? What is serializability of schedule in Transaction process? (10)
6. What are Lock based protocols in DBMS? Explain. (10)

2

MCA-6102

SECTION - D

7. Define XML database? Compare and contrast XML enable and native Xml database? (10)
8. Explain data formats and spatial relation with reference to Geographic Information System? (10)

SECTION - E (Compulsory Questions)

9. Explain the following:
 - (a) Explain data anomalies.
 - (b) What is E-R diagram?
 - (c) What is DML and DCL?
 - (d) Explain data encryption.
 - (e) BCNF?
 - (f) Components of DBMS?
 - (g) What is INDEX?
 - (h) Explain Transaction transparency.
 - (i) What is shadow paging?
 - (j) What is functional dependency? (10×2=20)